

Alex M. Chubaty, Ph.D.

Ecologist and Simulation Modeller

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EDUCATION

PhD, Biological Sciences, Simon Fraser University, Burnaby, BC 2014

BSc, Biological Sciences, University of Calgary, Calgary, AB 2005

WORK EXPERIENCE

President

- FOR-CAST Research & Analytics Oct 2019 – present
 - Ecological modelling and scientific software development.
 - Selected projects:
 - Historic and future range of variation of British Columbia forest landscapes, simulated with LANDIS-II (Biomass Succession with the Dynamic Fire and Dynamic Fuel systems), with fire regimes calibrated to observed fire perimeters and ignitions (2022–present);
 - Old-forest connectivity mapping with circuit-theory models (Omniscape) to inform biodiversity management in the Quesnel Natural Resource District, BC, for Phase 3 of the Connectivity, Corridors, and Functional Ecological Networks (C-C-FEN) project (2025–present);
 - Mountain pine beetle recruitment rates (r values) in Alberta: data retrieval and analyses supporting Cooke et al. (2026) (2024–present);
 - Spruce budworm atmospheric transport model: individual-based simulations of long-distance moth dispersal, reported on a public project website (2025–present);
 - Forest tent caterpillar outbreaks across Canada, 1938–2001: synchrony in time and spatial pattern in relation to forest and climate variables (2023–present);
 - Forest defoliators in northwestern Ontario: joint modelling of defoliator outbreaks and host trees (2026–present);
 - Integrated forest pest management: pine host layers from satellite-based forest inventory data (2025–present).

Spatial Modelling Coordinator

- fRI Research June 2018 – Sept 2019
 - Development and maintenance of the LandWeb Natural Range of Variation simulation model;
 - Development and maintenance of the LandWeb web interface (“app”);
 - Development of spatial spread models of mountain pine beetle (MPB) and fire interactions in AB/SK.

Postdoctoral Researcher

- Canadian Forest Service — Pacific Forestry Centre Sept 2015 – Mar 2018
 - Integrate conceptual models of mountain pine beetle (MPB) dispersal and available data into simulation models to predict the easterly and northerly spread of MPB in Canada.

- Develop spatial simulation software: SpaDES <https://spades.predictiveecology.org/>
- Université Laval — Département des sciences du bois et de la forêt AND
Canadian Forest Service — Pacific Forestry Centre Mar 2014 – Mar 2018
 - Development of individual based models of mountain pine beetle dispersal on the landscape.
 - Development of spatial simulation software, SpaDES <https://spades.predictiveecology.org/>

Research Assistant

- Simon Fraser University — Surrey Campus Oct 2012 – May 2013
 - Developer and editor of an online journal publishing research essays by undergraduate students.
 - Grant awarded by SFU Teaching and Learning Development Centre to Dr. Ivona Mladenovic.

Modeller

- Canadian Forest Service — Northern Forestry Centre Jan 2007 – Mar 2007
 - Continued development of a model evaluating mountain pine beetle host-acceptance and aggregation

Lab Assistant

University of Calgary

- Sikes Lab Nov 2004 – Aug 2005
 - Insect collection maintenance; miscellaneous lab work
- Longair Lab Sept 2004 – Aug 2005
 - Sorting, identifying, pinning insect specimens

PUBLICATIONS

* Equal contribution.

- Cooke, B.J., **A.M. Chubaty**, A.L. Carroll (2026) The mountain pine beetle in a marginal boreal landscape: Cross-scale collapse triggered by population removal. *Forest Ecology and Management* 609:123645. [10.1016/j.foreco.2026.123645](https://doi.org/10.1016/j.foreco.2026.123645).
- Raymundo, A., T. Micheletti, S. Haché, D. Stralberg, F.E.C. Stewart, J.A. Tremblay, C. Barros, I.M.S. Eddy, **A.M. Chubaty**, M. Leblond, C.L. Mahon, S.L. Van Wilgenburg, E.M. Bayne, F. Schmiegelow, T.D.S. Docherty, E.J.B. McIntire, S.G. Cumming (2024) Climate-sensitive forecasts of marked short-term and long-term changes in the distributions or abundances of Northwestern boreal landbirds. *Climate Change Ecology* 7:100079. [10.1016/j.ecochg.2023.100079](https://doi.org/10.1016/j.ecochg.2023.100079).
- Stewart, F.E.C., T. Micheletti, S.G. Cumming, C. Barros, **A.M. Chubaty**, A.L. Dookie, I. Duclos, I. Eddy, S. Haché, J. Hodson, J. Hughes, C.A. Johnson, M. Leblond, F.K.A. Schmiegelow, J.A. Tremblay, E.J.B. McIntire (2023) Climate-informed forecasts reveal dramatic local habitat shifts and population uncertainty for northern boreal caribou. *Ecological Applications* 33(3):e2816. [10.1002/eap.2816](https://doi.org/10.1002/eap.2816).
- Micheletti, T., S. Haché, D. Stralberg, F.E.C. Stewart, **A.M. Chubaty**, C. Barros, E.M. Bayne, S.G. Cumming, T.D.S. Docherty, A. Dookie, I. Duclos, I.M.S. Eddy, Z. Gadallah, C.A. Haas, J. Hodson, M. Leblond, C.L. Mahon, F. Schmiegelow, J.A. Tremblay, S.L. Van Wilgenburg, A.R. Westwood, E.J.B. McIntire (2023) Will

this umbrella leak? A caribou umbrella index for boreal landbird conservation. *Conservation Science and Practice* 5(4):e12908. [10.1111/csp2.12908](https://doi.org/10.1111/csp2.12908).

- Barros, C., Y. Luo, **A.M. Chubaty**, I.M.S. Eddy, T. Micheletti, C. Boisvenue, D.W. Anderson, S.G. Cumming, and E.J.B. McIntire (2023) Empowering ecological modellers with a PERFICT workflow: Seamlessly linking data, parameterisation, prediction, validation and visualisation. *Methods in Ecology and Evolution* 14(1):173–188. [10.1111/2041-210X.14034](https://doi.org/10.1111/2041-210X.14034).
- Boisvenue, C., G. Paradis, I.M.S. Eddy, E.J.B. McIntire, **A.M. Chubaty** (2022) Managing forest carbon and landscape capacities. *Environmental Research Letters* 17(11):114013. [10.1088/1748-9326/ac9919](https://doi.org/10.1088/1748-9326/ac9919).
- McIntire, E.J.B., **A.M. Chubaty**, S.G. Cumming, D. Anderson, C. Barros, C. Boisvenue, S. Haché, Y. Luo, T. Micheletti, F.E.C. Stewart (2022) PERFICT: A re-imagined foundation for predictive ecology. *Ecology Letters* 25(6):1345–1351. [10.1111/ele.13994](https://doi.org/10.1111/ele.13994).
- Micheletti, T., F.E.C. Stewart, S.G. Cumming, S. Haché, D. Stralberg, J.A. Tremblay, C. Barros, I.M.S. Eddy, **A.M. Chubaty**, M. Leblond, R.F. Pankratz, C.L. Mahon, S.L. Van Wilgenburg, E.M. Bayne, F. Schmiegelow, E.J.B. McIntire (2021) Assessing Pathways of Climate Change Effects in SpaDES: An Application to Boreal Landbirds of Northwest Territories Canada. *Frontiers in Ecology and Evolution* 9:679673. [10.3389/fevo.2021.679673](https://doi.org/10.3389/fevo.2021.679673).
- **Chubaty, A.M.**, P. Galpern, and S. Doctolero (2020) The R toolbox grainscape for modelling and visualizing landscape connectivity using spatially explicit networks. *Methods in Ecology and Evolution* 11(4):591–595. [10.1111/2041-210X.13350](https://doi.org/10.1111/2041-210X.13350).
- Bauduin, S., E.J.B. McIntire, **A.M. Chubaty** (2019) NetLogoR: A package to build and run spatially explicit agent-based models in R. *Ecography* 42(11):1841–1849. [10.1111/ecog.04516](https://doi.org/10.1111/ecog.04516).
- Zappia, S.P.W., **A.M. Chubaty**, B.D. Roitberg (2018) State-dependent domicile leaving rates in *Anopheles gambiae*. *Malaria Journal* 17:25. [10.1186/s12936-017-2166-4](https://doi.org/10.1186/s12936-017-2166-4).
- **Chubaty, A.M.**, M. Hart, B.D. Roitberg (2014) To tree or not to tree: The role of energy limitation on host tree acceptance in a bark beetle. *Evolutionary Ecology Research* 16(4):337–349.
- **Chubaty, A.M.***, B.O. Ma*, R.W. Stein, D.R. Gillespie, L.M. Henry, C. Phelan, E. Palsson, F.W. Simon, B.D. Roitberg (2014) On the evolution of omnivory in a community context. *Ecology and Evolution* 4:251–265. [10.1002/ece3.923](https://doi.org/10.1002/ece3.923)
- Major, H.L. and **A.M. Chubaty** (2012) Estimating colony and breeding population size for nocturnal burrow-nesting seabirds. *Marine Ecology Progress Series* 454:83–90. [10.3354/meps09663](https://doi.org/10.3354/meps09663)
- Jumeau, Z., B.O. Ma, **A.M. Chubaty**, C.W. Ellenor, B.D. Roitberg, and G. Gries (2011) A theoretical approach to study the evolution of aggregation behavior by larval codling moth, *Cydia pomonella* (Lepidoptera: Tortricidae). *Journal of Insect Behavior* 24(4):249–263. [10.1007/s10905-010-9251-7](https://doi.org/10.1007/s10905-010-9251-7)
- **Chubaty, A.M.**, B.D. Roitberg, C. Li (2009) A dynamic host selection model for mountain pine beetle, *Dendroctonus ponderosae* Hopkins. *Ecological Modelling* 220:1241–1250. [10.1016/j.ecolmodel.2009.01.039](https://doi.org/10.1016/j.ecolmodel.2009.01.039).

Software package development

Author

- AGBtrends: Utilities for analyzing trends in above ground biomass. [GitHub](#)
- bcwsft: BC Wildfire Service fuel typing algorithm (R port). [GitHub](#)
- BiomeBGCR: Run Biome-BGC models from R. [GitHub](#)
- CBMutils: Utilities for modelling carbon based on CBM-CFS3. [GitHub](#)
- climateData: Utilities for working with North American climate data. [GitHub](#)
- crew.ssh: 'crew' launcher plugin for SSH workers on a local network. [GitHub](#)
- fireregimetools: Tools for summarizing fire regime statistics from simulations. [GitHub](#)
- fireSenseUtils: Utilities for working with the 'fireSense' group of 'SpaDES' modules. [GitHub](#)
- forestData: Tools for importing Canadian forestry data. [GitHub](#)
- fpCompare: Reliable comparison of floating point numbers. [CRAN](#); [GitHub](#); [Website](#)
- grainscape: Landscape connectivity, habitat, and protected area networks. [CRAN](#); [GitHub](#); [Website](#)
- landisbc: British Columbia study-area tools for LANDIS-II workflows. [GitHub](#)
- landisutils: Utilities for preparing, running, and post-processing LANDIS-II simulations. [GitHub](#); [Website](#)
- LandR: Landscape Ecosystem Modelling in R. [GitHub](#); [Website](#)
- LandWebUtils: Additional utilities for LandWeb analyses. [GitHub](#); [Website](#)
- libcbmr: Interface to the Carbon Budget Model library based on CBM-CFS3. [GitHub](#)
- map: Defines a meta class of geographical objects, the 'map' class, and associated tools. [GitHub](#); [Website](#)
- mpbutils: Utilities for mountain pine beetle simulation modelling. [GitHub](#)
- NetLogoR: Build and run spatially explicit agent-based models. [CRAN](#); [GitHub](#); [Website](#)
- nrvtools: Tools for summarizing outputs from natural range of variability simulations. [GitHub](#)
- pemisc: Miscellaneous utilities developed by the Predictive Ecology Lab Group. [GitHub](#); [Website](#)
- peutils: Utilities developed by the Predictive Ecology Group. [GitHub](#); [Website](#)
- sbwutils: Utilities for modelling spruce budworm eruptions across Canada. [GitHub](#)
- scfmutils: Tools and utilities for working with the SCFM wildfire simulation model. [GitHub](#); [Website](#)
- SpaDES.targets: Orchestrate SpaDES simulations with 'targets'. [GitHub](#)
- spatialutils: Utilities for working with spatial data. [GitHub](#)
- workflowtools: Project workflow management tools and utilities. [GitHub](#)
- ytdata: Fetch Yukon open geospatial data. [GitHub](#)

SpaDES simulation platform (co-developer)

- quickPlot: A system of plotting optimized for speed and modularity. [CRAN](#); [GitHub](#); [Website](#)
- reproducible: Enhance reproducibility of R code. [CRAN](#); [GitHub](#); [Website](#)
- SpaDES: Metapackage for the SpaDES family. [GitHub](#); [Website](#)
- SpaDES.addins: Tools and RStudio addins for 'SpaDES' module development. [GitHub](#)
- SpaDES.core: Core utilities for developing and running spatially explicit discrete event models. [CRAN](#); [GitHub](#); [Website](#)
- SpaDES.docs: Helper utilities for creating SpaDES module documentation and manuals. [GitHub](#); [Website](#)
- SpaDES.experiment: Simulation experiments within the SpaDES ecosystem. [GitHub](#); [Website](#)
- SpaDES.tools: Additional tools for developing spatially explicit discrete event simulation (SpaDES) models. [CRAN](#); [GitHub](#); [Website](#)

Contributor

- BioSIM: Client for the BioSIM web API. [GitHub](#)
- J4R: Create 'Java' objects and execute 'Java' methods. [GitHub](#)
- LandR.CS: Climate-sensitive growth and mortality in LandR. [GitHub](#)
- lazyR: Create lazy load databases for R objects. [GitHub](#)
- miniCRAN: Create a mini version of CRAN containing only selected packages. [CRAN](#); [GitHub](#); [Website](#)
- PSPclean: Sources and standardizes permanent sample plot data. [GitHub](#)
- Require: Installing and loading R packages for reproducible workflows. [CRAN](#); [GitHub](#); [Website](#)
- SpaDES.project: Project templates using 'SpaDES'. [GitHub](#); [Website](#)
- tiler: Create geographic and non-geographic map tiles. [CRAN](#); [GitHub](#); [Website](#)

LANDIS-II contributions

- LANDIS-II Docker and Apptainer images: containers with LANDIS-II and its extensions, for workstations and computing clusters. [GitHub](#)
- LANDIS-II extensions: build fixes for 13 extensions (library references, including the Universal Cohort Library v2 update), a bug fix in ForCS Succession, and documentation updates for Base Fire and Biomass Succession. [GitHub](#)
- LANDIS-II Forest Product Sector extension: carbon transfers and emissions from harvested wood products. [GitHub](#)

Other software experience

- Operating Systems: Windows, macOS, Ubuntu Linux
- Office productivity software: Microsoft Office (Word, Excel and PowerPoint), Open Office, LaTeX
- Online Course Management Software: Blackboard, WebCT
- Programming languages: C/C++, R
- Simulation models: LANDIS-II, SpaDES
- Landscape connectivity: grainscape, Omniscape
- Reproducible workflows: targets, renv, Quarto, R Markdown, Docker, Apptainer
- High Performance Computing: Compute Canada (now Digital Research Alliance of Canada), HTCondor
- Version control systems: git, GitHub
- Web development: CSS, HTML, Open Journal Systems, shiny

PRESENTATIONS

- **Chubaty, A. M.**, and E. J. B. McIntire. A brief introduction to SpaDES. May 8, 2020.
- Barros, C., **A. M. Chubaty**, S. G. Cumming, T. Micheletti, I. Eddy, C. Boisvenue, F. E. C. Stewart, and E. J. B. McIntire. Making Predictive Ecology PERFICT. Belfast, Northern Ireland. Dec 11, 2019.
- **Chubaty, A. M.**, T. Micheletti, F. Stewart, E. J. B. McIntire, Y. Luo, D. W. Andison, S. Haché, I. Eddy, C. Barros, D. Stralberg, J. Marchal, A. P. Raymundo, M. van Telgen, J. Tremblay, M. Leblond, R. Pankratz, J. Hodson, and S. Cumming. Cumulative effects research in SpaDES. Sault Ste Marie, ON. Oct 25, 2019.
- Micheletti, T., E.J.B. McIntire, F. Stewart, S. Haché, I. Eddy, C. Barros, **A.M. Chubaty**, D. Stralberg, J. Marchal, A.P. Raymundo, M. van Telgen, J. Tremblay, M. Leblond, R. Pankratz, J. Hodson, S.G. Cumming. Data to decisions: a multispecies approach case study for Northwest Territories. Society of Canadian Ornithologists Meeting, Aug 30, 2019.
- Stewart, F., T. Micheletti, E.J.B. McIntire, S. Haché, M. Leblond, J. Tremblay, J. Marchal, M. van Telgen, J. Hodson, F. Schmiegelow, I. Eddy, **A.M. Chubaty**, C. Barros, S.G. Cumming. Predicting caribou resource selection and demography under landscape and climate change. Canadian Society for Ecology and Evolution

Meeting, Aug 20, 2019.

- **Chubaty, A.M.** Best Practices Working With Spatial Data in R. fRI Research, May 2, 2019
- **Chubaty, A.M.**, E.J.B. McIntire, B.J. Cooke. Facing the tree-slayer: forecasting Mountain Pine Beetle spatial spread in SpaDES. fRI Research Mountain Pine Beetle Ecology Program Research Flash, Feb 22, 2019.
- **Chubaty, A.M.**, E.J.B. McIntire, B.J. Cooke. Integrating fire disturbance with models of mountain pine beetle spread to evaluate efficacy of controlled burns as a management tool. SERG International Workshop, 5–7 Feb 2019.
- **Chubaty, A.M.** Testimony before the House of Commons Standing Committee on Natural Resources Forest Pests. Oct 4, 2018.
- **Chubaty, A.M.**, E.J.B. McIntire. Cumulative Impacts Monitoring Program (CIMP) SpaDES Workshop. Cumulative Impacts Monitoring Program, Oct 3–4, 2018.
- **Chubaty, A.M.**, E.J.B. McIntire, D.W. Anderson. LandWeb: Exploring NRV in the Western Canadian Boreal Forest. fRI Research, July 27, 2018.
- **Chubaty, A.M.** Building R packages - Getting started and best practices. University of Calgary Biological Sciences R Users Group, May 11, 2018.
- **Chubaty, A.M.**, B.J. Cooke, E.J.B. McIntire. Simulating MPB invasive spread control in Saskatchewan using SpaDES. SERG International Workshop, 6–8 Feb 2018.
- McIntire, E.J.B., S. Bauduin, J. Marchal, A. Clason, **A.M. Chubaty**, S.G. Cumming. Predictive ecology in a world filled with dynamic data and models – entering continuous adaptive management. University of Victoria Environmental Studies Seminar Series, Jan 2018.
- McIntire, E.J.B., S. Bauduin, J. Marchal, A. Clason, **A.M. Chubaty**, S.G. Cumming. Incorporating changing climate in ecological models: feedbacks, trade-offs and surprises! OR How do we influence decision makers? Canadian Institute of Forestry, Web Seminar, Nov 2017.
- **Chubaty, A.M.**, E.J.B. McIntire, B.J. Cooke. Boreal insect disturbance in SpaDES: integrated simulation model of mountain pine beetle eastward spread. Joint Annual Meeting of the Entomological Societies of Canada and Manitoba, 22–25 Oct 2017.
- **Chubaty A.M.** and E.J.B. McIntire. Cumulative effects simulation and modern scientific forecasts: challenges and opportunities. Great Lakes Forestry Centre Seminar Series, NRCAN, 20 Oct 2017.
- McIntire, E.J.B., S. Bauduin, J. Marchal, A. Clason, **A.M. Chubaty**, S.G. Cumming. Incorporating changing climate in ecological models: feedback, reversals and surprises abound! Global climate change and patagonian ecosystem response: present and future challenges, Coyhaique, Chile, 2017.
- McIntire, E.J.B., E.M. Campbell, R.S. Winder, **A.M. Chubaty**, M.A. Wulder, *et al.* Building a continuous adaptive management system in SpaDES: dialing down the risk for woodland caribou. Ontario, 2017.
- **Chubaty, A.M.**, E.J.B. McIntire, B.J. Cooke, S.G. Cumming, Y. Luo. Boreal insect disturbance in SpaDES: integrated simulation model of mountain pine beetle eastward spread (poster). Canadian Society for Ecology and Evolution, 7–10 May 2017.
- Cumming, S.G., E.J.B. McIntire, Y. Luo, **A.M. Chubaty**, D. Woolford. Burning in SpaDES: Automated parameter estimation for landscape fire models. Canadian Society for Ecology and Evolution, 7–10 May 2017.
- Luo, Y., E.J.B. McIntire, **A.M. Chubaty**, D. Anderson, S.G. Cumming. Vegetation dynamics in SpaDES: the recoded LANDIS-II biomass succession model. Canadian Society for Ecology and Evolution, 7–10 May 2017.
- McIntire, E.J.B., D. Anderson, S.G. Cumming, Y. Luo, **A.M. Chubaty**. Building a continuous adaptive management system in SpaDES: using historical landscape variation for the LandWeb project. Canadian Society for Ecology and Evolution, 7–10 May 2017.
- **Chubaty, A.M.**, E.J.B. McIntire, B.J. Cooke. Simulating MPB invasive spread control using SpaDES. fRI Mountain Pine Beetle Ecology Research Forum, 25–26 Apr 2017.
- **Chubaty, A.M.**, E.J.B. McIntire, B.J. Cooke. Simulating MPB invasive spread control in Saskatchewan using SpaDES. SERG International Workshop, 6–9 Feb 2017.

- McIntire, E.J.B., **A.M. Chubaty**, D. Andison. Getting to answers, in Spades. FaCOP Webinar Series, Sault Ste. Marie, ON, 2016.
- McIntire, E.J.B. and **A.M. Chubaty**. The Spades platform: Modern scientific forecasts, backcasts, and decisions. Northern Forestry Centre, NRCAN, 22 July 2015.
- Riel, W., **A.M. Chubaty**. Evolution of Decision Support for Forest Ecosystem Management: Towards Open Modelling and Data. Annual General Meeting of the Entomological Society of British Columbia (Symposium), 17 Oct 2015.
- **Chubaty, A.M.** and E.J.B. McIntire. Spades: Spatial Discrete Event Simulation. Northern Forestry Centre, NRCAN. 25 Nov 2014.
- Hodson, C., S. Dhanani, A. Hoi, **A.M. Chubaty**, F. Simon. *Drosophila suzukii* in a *D. suzukii* world: Humidity decreases density dependent competition. Annual General Meeting of the Entomological Society of British Columbia, 1–2 Nov 2013
- **Chubaty, A.M.**, B.O. Ma, R.W. Stein, D.R. Gillespie, L.M. Henry, C. Phelan, E. Palsson, F.W. Simon, B.D. Roitberg. On the evolution of omnivory in a community context. 98th Annual Meeting of the Ecological Society of America, 4–9 Aug 2013.
- Mladenovic, I., **A.M. Chubaty**, and A. Tirajoh. Enhancing Students' Understanding of Biological Concepts by Engaging Them in Research (poster). Symposium on Teaching and Learning: Embracing Change @ SFU, 15 May 2013.
- **Chubaty, A.M.**, B.O. Ma, R.W. Stein, D.R. Gillespie, L.M. Henry, C. Phelan, E. Palsson, F.W. Simon, B.D. Roitberg. On the evolution of omnivory in a community context. Canadian Society for Ecology and Evolution, 12–15 May 2013.
- **Chubaty, A.M.**, B.O. Ma, R.W. Stein, D.R. Gillespie, L.M. Henry, C. Phelan, E. Palsson, F.W. Simon, B.D. Roitberg. Evolution of omnivory in a community context. RPM Research Group, SFU, 11 Apr 2013.
- **Chubaty, A.M.** To tree or not to tree: Energy and time limited habitat settlement decisions in bark beetles. Les Écologistes Seminar Series, SFU, Nov 2012.
- **Chubaty, A.M.** and B.D. Roitberg. Can selfish genes help prevent pest outbreaks? 1st Joint Congress on Evolutionary Biology, 6–10 July 2012.
- Roitberg, B.D. and **A.M. Chubaty**. State dependent problems for arthropods. Pacific Institute for the Mathematical Sciences / Centre for Scientific Computing Seminar Series, SFU, 19 Nov 2010.
- **Chubaty, A.M.** Energy reserves modulate host acceptance decisions in mountain pine beetles. Joint Annual Meeting of the Entomological Societies of Canada and British Columbia, 31 Oct – 3 Nov 2010.
- **Chubaty, A.M.** and M. Hart. Mountain pine beetle condition and timing of emergence: who emerges when. Annual General Meeting of the Entomological Society of British Columbia, 2–3 Oct 2009. (*Best PhD Student Talk*)
- **Chubaty, A.M.** and M. Hart. Mountain pine beetle condition and timing of emergence: who emerges when. Canadian Society for Ecology and Evolution, 11–14 May 2008.
- **Chubaty, A.M.**, B.D. Roitberg, C. Li. An individual model of host selection for mountain pine beetle. Canadian Society for Ecology and Evolution, 17–20 May 2007.
- **Chubaty, A.M.**, *et al.* "I'll have the steak special, with a side salad": exploring the evolution of omnivory. Les Écologistes Seminar Series, SFU, 15 Mar 2007.
- **Chubaty, A.M.**, B.D. Roitberg, C. Li. Beetles pine for lodgepole homes: mountain pine beetle's search for a host. Annual General Meeting of the Entomological Society of British Columbia, 13–14 Oct 2006.
- **Chubaty, A.M.**, B.D. Roitberg, C. Li, B.O. Ma. Using game theory to evaluate host acceptance decisions in mountain pine beetle. 27th Annual Pacific Ecology and Evolution Conference, 3–5 Mar 2006.

TEACHING EXPERIENCE

Sessional Instructor

Simon Fraser University

- BISC 300 Evolution Fall 2013
 - Taught lectures; marked assignments and exams
 - Topics covered include: adaptation, speciation, origin of life, evolutionary trends over geological time, origin and evolution of modern humans
- BISC 407 Population Dynamics Spring 2011
 - Taught lecture and tutorial sessions; marked assignments and exams
 - Evaluated students' presentations of scientific papers and led discussions on population modelling topics
 - Topics covered include: lifetables, survivorship, population growth models, differential solvers, intraspecific competition, demographics, population matrices, interspecific competition, community structure, predation, parasitoid/host dynamics, metapopulation dynamics, infectious diseases

Guest Lecturer

Simon Fraser University

- BISC 888-1 Directed Readings: Data Wrangling and Visualization in R Fall 2013
 - **Graduate course**
 - Topics covered: introduction to R and markdown; graphics in R (base plot and ggplot); data manipulation in R (plyr and reshape)
- BISC 100 Introduction to Biology Oct 2010, Nov 2010
 - Introduced students to cell metabolism and phenotype expression
- BISC 204 Introduction to Ecology Sept 2006, Nov 2006, Mar 2010, Nov 2011
 - Topics covered include: organism distribution and abundance and the theoretical and practical aspects of habitat selection in animals
 - Presented development of my models of individual behaviour and population dynamics in mountain pine beetle, and implications for landscape-level development
- BISC 410 Behavioural Ecology Feb 2008
 - Discussed theoretical and practical aspects of habitat selection in animals

Workshop Instructor

- Workshops in R:
 - Introduction to SpaDES (Pacific Forestry Centre, NRCAN) Sept 2016, Dec 2016
 - Introduction to R (BISC 838 Population Ecology, SFU) Sept 2010, Sept 2012
 - Introduction to R (Climate Change and Community Structure – NSERC Strategic Grant, SFU) Nov 2009
 - Introduction to R (Roitberg Lab Group, SFU) Sept 2006

Teaching Assistant

Simon Fraser University

- BISC 100 Introduction to Biology - Distance Education Fall 2011, Summer 2012
 - Taught general biological principles including cell functions, genetics, evolution, behaviour, population and community dynamics
 - Ran online tutorial discussions and exam review sessions; assisted with management and instruction of labs; marked assignments and exams
- BISC 100 Introduction to Biology Fall 2010, Spring 2012
 - Taught general biological principles including cell functions, genetics, evolution, behaviour, population and community dynamics
 - Designed and taught lessons for tutorial sessions; assisted with management and instruction of labs; marked assignments and exams
- BISC 102 Introductory Biology II Spring 2006, Spring 2009
 - Taught general biological principles including genetics, diversity and evolution, behaviour, population and community dynamics
 - Designed and taught lessons for tutorial sessions; assisted with management and instruction of labs; marked assignments and exams
- BISC 471W Special Topics: Biology Before the Beagle Fall 2008
 - Examination of the history of biological thought, emphasizing the life and times of Charles Darwin
 - Exploration of popular and technical scientific writing styles; writing for all audiences; writing historical science essays
 - Designed and taught lessons for tutorial sessions; marked essay assignments
- BISC 410 Behavioural Ecology Spring 2008
 - Topics covered: foraging, predation, aggression, habitat selection, reproduction, parental care, social behaviour, conservation
 - Evaluated students' presentations of scientific papers and led discussions on behavioural and ecological topics; marked assignments and exams
- BISC 204 Introduction to Ecology Fall 2006
 - Topics covered: evolution, life history, behavioural ecology, population and community ecology
 - Evaluated students' presentations of scientific papers and led discussions on ecological topics; marked assignments and exams

University of Calgary

- BIOL 311 Introductory Genetics Fall 2004
 - Topics covered include: Mendelian inheritance, linkage, recombination, gene expression
 - Assisted with management and instruction of labs; marked assignments and exams

Co-op Report Grader

- Simon Fraser University Co-op Program Summer 2011, Fall 2011

- Read, evaluated, and provided feedback on work-term reports submitted by co-op students

AWARDS, SCHOLARSHIPS & GRANTS

• Simon Fraser University Dean Graduate Studies Committee Travel Award (\$500)	Summer 2013
• Graduate Student Scholarship, Entomological Society of BC (\$400)	Oct 2012
• Simon Fraser University Dean Graduate Studies Committee Travel Award (\$500)	Summer 2012
• Simon Fraser University President's Research Stipend (\$6250)	Spring 2011
• Best PhD Student Presentation, Entomological Society of BC AGM (\$300)	Oct 2009
• Simon Fraser University Graduate Fellowship (\$6250)	Fall 2007, 2009
• Simon Fraser University Dean Graduate Studies Committee Travel Award (\$500)	Summer 2007
• Jason Lang Scholarship (\$1000)	Jan 2005
• University of Calgary Undergraduate Academic Scholarship (\$500)	Jan 2003
• H&R Block Scholarship (USD \$550)	Fall 2002, Winter 2003
• Jason Lang Scholarship (\$1000)	Dec 2002
• University of Calgary Undergraduate Merit Award (\$700)	Nov 2002
• John and Elsie Wawruch Memorial Scholarship (\$500)	Jan 2002
• H&R Block Scholarship (USD \$350)	Fall 2001, Winter 2002
• Alexander Rutherford Scholarship (\$2500)	Oct 2001
• University of Calgary Scholar's Advantage Award (\$500)	Sept 2001

MEMBERSHIP

• Canadian Society for Ecology and Evolution	2006–present
• Ecological Society of America	2013–2014
• Entomological Society of British Columbia	2005–2015
• Entomological Society of Canada	2010–2014; 2016–present
• Professional Pest Management Association of British Columbia	2007–2010

SERVICE ACTIVITY

• Reviewer, PeerJ	2017
• External Reviewer, NSERC discovery grants	2016
• Website Editor, Entomological Society of British Columbia	2011–2015
– Oversaw digitization of the society's entire journal archive dating back to 1906, and directed efforts to move the journal archive online and made freely available as an open-access publication.	
• A/V Volunteer, 98th Annual Meeting of the Ecological Society of America	4–9 Aug 2013
• Organizer, Joint Annual Meeting of the Entomological Societies of Canada and British Columbia	2010
• Volunteer, Let's Talk Science	2008–2013
• Organizer, SFU New Biology Grad Student Mentorship Program	2007–2010
• Webmaster, SFU Biology Grad Student Caucus	2007–2012
• Editor, Newsletter of the Professional Pest Management Association of British Columbia	2007–2010
• Webmaster, Professional Pest Management Association of British Columbia	2007–2010
• Judge, Calgary Youth Science Fair	2003, 2004
• Student Leader, "U of C 101" University of Calgary	2003, 2004